Twenty-fifth Section.

Transcendent Numbers.

§. 203.

Countable Sets.

In the introduction to our work the general concept of numbers was defined, and with this general concept of number we have at first operated, for example in the proof of the existence of roots. In the further course of our investigations we have then occupied ourselves only with algebraic numbers, without giving account to ourselves whether the extent of the number-domain was thereby exhausted, or whether there are also non-algebraic numbers. The existence of non-algebraic numbers, which are also called transcendent numbers, was first proved by Liouville. Other proofs of this theorem are due to G. Cantor[1].

We attach ourselves here to the concept, already set forth in the introduction, of a set or manifold; by this is to be understood a system of elements of any kind whatever, which is so precisely delimited that of every arbitrary object it is fully decided whether it belongs to the system or not.

We distinguish finite and infinite sets, and as the first and most important example of an infinite set we cite the totality of the natural numbers $1, 2, 3, \ldots$. Then the following definition holds:

1. Definition. A set is called countable if its elements can be put, with the natural number sequence or with a part of the same, into a reciprocal one-to-one relation[2].

Every finite set is accordingly countable, and in the counting the natural number sequence is used only up to a certain greatest number. In what follows we consider preferably infinite sets.

We can also give the definition for an infinite countable set the expression that it is such a set in which to each element a definite number of the natural number sequence can be assigned as name, in such a way that every number of the number sequence is also used, hence a set in which there is a first, second, third ... hundredth ... element.

Finally we can also say that an infinite countable set is one which can be ordered into a sequence in such a way that a first element is present, and that after each element a definite other element of the set follows, and before each element, with the exception of the first, a definite other element precedes. Such a sequence we can call a countable ordering.

It is clear that a countable set can be counted not only in one way, but in infinitely many different ways.

Besides the natural number sequence itself, which is certainly countable, we can cite as a second example the system of the rational positive proper fractions, which among other ways can be counted in the following way:

$$\frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{3}{4}, \frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}, \frac{1}{6}, \frac{5}{6}, \ldots$$

that is, in such a way that every larger denominator follows the smaller denominator, and that the fractions with the same denominator are ordered according to the size of the numerator. But if one wanted to order the fractions according to the size of their numerical value, one would not obtain a countable ordering.

2. The totality of all algebraic numbers is a countable set.

[1] G. Cantor, Crelle's Journal, vol. 77 (1873). Ueber eine Eigenschaft des Inbegriffes aller reellen algebraischen Zahlen.

[2] Cantor, loc. cit. The concept of countable sets coincides with Dedekind's concept, defined without presupposing the number-system, of simply infinite systems. (Dedekind, Was sind und was sollen die Zahlen? §. 6. Braunschweig 1887. Second unchanged edition 1893.)

To prove this important theorem, we recall that every algebraic number Θ is the root of one and only one irreducible equation

$$f(\Theta) = a_0\Theta^n + a_1\Theta^{n-1} + \cdots + a_n = 0 \tag{1}$$

in which $a_0, a_1, \ldots, a_n$ are rational integers without a common divisor, and a_0 is different from zero and positive. The degree n of the equation (1) is a positive integer, therefore at least $= 1$. The numbers $a_1, \ldots, a_{n-1}$ can in part be zero.

We shall now determine the signs $\pm$ so that $\pm a_1, \pm a_2, \pm a_3, \ldots, \pm a_n$ are not negative, and call the sum

$$N = (n-1) + a_0 \pm a_1 \pm a_2 \cdots \pm a_n \tag{2}$$

the height of the algebraic number Θ. The height is then always a positive integer.

Now it is easy to see that for a given value of the height N there can always be only a finite number of algebraic numbers. For first, by (2), n can never be greater than N, and for given N and n the numbers $a_0, a_1, \ldots, a_n$ can be determined only in a finite number of ways. Among the functions $f(\Theta)$ so determined one then has to retain only the irreducible ones. If one now orders the algebraic numbers in such a way that one puts the numbers of smaller height before those of greater height, that among numbers of equal height one puts first those whose real part is smaller, and among numbers of equal height and equal real part lets those with smaller imaginary part precede, then we have a countable ordering of the algebraic numbers, and it is proved that the totality of all algebraic numbers is a countable set.

For example, we obtain:

$$\begin{array}{lllll}
N = 1, & n = 1, & a_0 = 1, & a_1 = 0, & \\
N = 2, & n = 1, & a_0 = 1, & a_1 = \pm 1, & \\
N = 3, & n = 1, & a_0 = 1, & a_1 = \pm 2, & \\
 & & a_0 = 2, & a_1 = \pm 1, & \\
 & n = 2, & a_0 = 1, & a_1 = 0, & a_2 = 1,
\end{array}$$

and the beginning of the ordered sequence of algebraic numbers is therefore:

$$0,\ -1,\ +1,\ -2,\ -\frac{1}{2},\ -\sqrt{-1},\ \sqrt{-1},\ \frac{1}{2},\ 2, \ldots.$$

Every part of the natural number sequence is a countable set; for one need only order the numbers of such a part according to their size in order to obtain a countable ordering.

From this, however, it follows that every part of a countable set is itself a countable set. It follows from this, among other things, that the set of real algebraic numbers is also countable.

§. 204.

Uncountable Sets.

We now come secondly to the proof of the theorem that among sets of numbers there are also those which are not countable, and we shall prove specifically:

That the totality of all real numbers, even if we restrict ourselves to a finite interval, is not countable.

For this purpose we consider any countable set of real numbers distinct from one another, which, put into a countable ordering Ω, we shall denote as

$$(\Omega) \qquad \omega_1, \omega_2, \omega_3, \omega_4, \ldots$$

(where it is to be remembered that this is not meant to be a succession according to size).

For brevity we shall call, of two elements of the sequence Ω, the one with the smaller index the earlier, and the one with the greater index the later.

We now take any two real numbers α, β, so that $\alpha < \beta$, and show that in the interval $\delta = (\alpha, \beta)$ there is at least one number which does not occur in the sequence Ω. If we have proved such a number for every interval, then there are also infinitely many of them, since the same inference can be applied to every part of the interval (α, β).

First it is clear that our assertion is correct if in some finite part of the interval δ only a finite number of numbers of the sequence Ω is contained, and we can therefore pass at once to the case in which in every, however small, part of the interval δ there lies an infinite set of numbers of the sequence Ω.

We denote by α_1, β_1 the two earliest numbers of the sequence Ω which lie in the interval δ, assume $\alpha_1 < \beta_1$, and set $\delta_1 = \beta_1 - \alpha_1$, so that $\delta_1 = (\alpha_1, \beta_1)$ is a part of the interval δ.

We denote in the same way by α_2, β_2 the two earliest numbers of Ω which lie in the interior of the interval δ_1 (excluding the endpoints), assume $\alpha_2 < \beta_2$, and set $\delta_2 = \beta_2 - \alpha_2$.

In this way we can continue and obtain an unbounded sequence of intervals

$$\delta,\ \delta_1,\ \delta_2,\ \delta_3, \dots,$$

each of which includes all the following ones, and two sequences of numbers

$$\alpha, \alpha_1, \alpha_2, \dots, \qquad \beta, \beta_1, \beta_2, \dots$$

which, with the possible exception of the first, α and β, all belong to the sequence Ω. The $\alpha, \alpha_1, \alpha_2, \dots$ form an increasing, the $\beta, \beta_1, \beta_2, \dots$ a decreasing number-sequence, and at the same time every α is smaller than every β.

1. From this it follows that the numbers α_ν have an upper bound a, the numbers β_ν a lower bound b, and that a in any case is not greater than b. (Compare vol. I, §. 38, 1.)

But possibly $a = b$.

From the way in which the intervals $\delta, \delta_1, \delta_2, \dots$ are formed, the following also emerges:

2. If any number ω of the sequence Ω lies in the interval δ_ν, excluding its endpoints α_ν, β_ν, then ω is later in the sequence Ω than the number-pair α_ν, β_ν belonging to that same sequence.

For α_ν, β_ν were precisely the two earliest numbers of Ω lying in the interval $\delta_{\nu-1}$. From this it follows:

3. The number-pairs α_ν, β_ν belonging to the sequence Ω are later members of the sequence Ω the larger the index ν is, and since we have assumed that the sequence of intervals δ_ν does not break off, we can for sufficiently large ν push α_ν, β_ν arbitrarily far out in the sequence Ω.

It now follows very simply that no number g which coincides with one of the numbers a, b, or also, if a and b are different, lies between them, can belong to the sequence Ω.

For the number g lies in the interior of each of the intervals δ_ν. Suppose that g occurs in Ω, and by 3. go so far with ν that α_ν, β_ν occur in Ω later than g; then, by 2., g cannot lie in the interval δ_ν, and thereby the impossibility of our assumption is shown.

Thus it follows that the totality of the numbers of an interval α, β does not form a countable set.

This fact can also be proved in another way, which in a certain respect is still simpler and which we shall present in few words. We do not restrict the generality if we take as basis the interval from 0 to 1. We think of all numbers of this interval as represented by infinite decimal fractions. Among them the finite decimal fractions are also contained, if from a certain place on we put all digits equal to zero. In order to make the representation by decimal fractions unique, let it still be fixed that for a finite decimal fraction this representation is always chosen, hence for example that 0,4999... is not to be put for 0,5000....

We now want to assume that these decimal fractions form a countable set; they can therefore be ordered in a countable sequence, which we write as follows:

$$(\Omega)\qquad \begin{array}{rcl} \omega_1 &=& 0,\alpha_1^{(1)}\alpha_2^{(1)}\alpha_3^{(1)}\dots \\ \omega_2 &=& 0,\alpha_1^{(2)}\alpha_2^{(2)}\alpha_3^{(2)}\dots \\ \omega_3 &=& 0,\alpha_1^{(3)}\alpha_2^{(3)}\alpha_3^{(3)}\dots \\ && \dots \end{array}$$

where the $\alpha_\mu^{(\nu)}$ denote digits of the decimal system.

It is now very easy to exhibit a decimal fraction (or also arbitrarily many) which are not contained in the sequence Ω. We need only form

$$\eta = 0,\beta_1\beta_2\beta_3\dots,$$

where the β_ν are digits of the decimal system satisfying the single condition that, for every ν, β_ν is different from $\alpha_\nu^{(\nu)}$. This number η, which after all also belongs to the interval $(0,1)$, cannot agree with any number of the sequence Ω.

One can generalize the formation of η still further by taking the first β's, up to an arbitrarily far removed place, arbitrarily, and only from there on requiring the law: β_ν different from $\alpha_\nu^{(\nu)}$, to hold.

Since it has now been proved that the real algebraic numbers form a countable set, and that in every interval there are numbers which do not belong to a given countable set, it follows from this quite immediately:

In every real interval there are transcendent numbers.

§. 205.

Transcendence of the Number e.

A much more difficult task is now, for a definitely presented number, to decide whether it is algebraic or transcendent. Here interest has concentrated chiefly on the two numbers, so frequently occurring in analysis, e, that is, the base of the natural logarithm system, and Ludolph's number π, the ratio of the circumference of the circle to the diameter.

For the number e the question was decided by Hermite in a famous paper, which became the foundation for the later investigations on the number π[3]. The decision for the number π, which because of its relation to the ancient and famous problem of the quadrature of the circle was of quite special interest, still offered insurmountable difficulties for a long time. Finally Lindemann gave the proof that π too belongs to the transcendent numbers. The proof given by Lindemann, however, at first still offered great difficulties for understanding, which were removed by later investigations of Weierstrass, Hilbert, Hurwitz, and Gordan[4] The proof of Weierstrass has simplified the form of the proof quite considerably, and this proof is to be set forth in what follows.

[3]Hermite, Sur la fonction exponentielle. Comptes rendus, vol. LXXVII, 1873.

[4]Lindemann, Ueber die Zahl π. Mathem. Annalen, vol. 20, 1882. Weierstrass, Zu Lindemann's Abhandlung "Ueber die Ludolph'sche Zahl". Sitzungsbericht der Berliner Akademie, 3 December 1885. The works of Hilbert, Hurwitz, and Gordan are all three found in volume 43 of the Mathematische Annalen (1893), the first two also in the Göttinger Nachrichten of 1893.

We denote the product of the natural numbers from 1 to n by

$$\Pi(n) = 1.2.3 \cdots n.$$

If x is an arbitrary real or complex number, then it can first be shown that, for an arbitrarily given value of x, the terms of the sequence

$$\frac{x^n}{\Pi(n)} \tag{1}$$

approach the limit zero as n grows without bound, and indeed more strongly than the terms of a decreasing geometric progression. For if k is an integer greater than the absolute value of x, then

$$\left|\frac{x^n}{\Pi(n)}\right| = \left|\frac{x^k}{\Pi(k)}\frac{x}{k+1}\frac{x}{k+2}\cdots\frac{x}{n}\right| < \left|\frac{x^k}{\Pi(k)}\right|\left|\frac{x}{k}\right|^{n-k}.$$

From this it follows that the infinite series

$$e^x = 1 + x + \frac{x^2}{\Pi(2)} + \frac{x^3}{\Pi(3)} + \cdots \tag{2}$$

converges for all values of x, and by it we define the exponential function e^x, whose simplest properties we here presuppose from analysis. In particular, for two arbitrary values x, y the relation

$$e^{x+y} = e^x e^y$$

holds, from which it then follows that e^n, for a positive integer n, is the nth power of the number

$$e = 2 + \frac{1}{\Pi(2)} + \frac{1}{\Pi(3)} + \frac{1}{\Pi(4)} + \cdots = 2{,}718281828459\ldots \tag{3}$$

If now r denotes any positive integer, we can write formula (2) in the form

$$\Pi(r)e^x = \Pi(r) + \frac{\Pi(r)}{\Pi(1)}x + \frac{\Pi(r)}{\Pi(2)}x^2 + \cdots + x^r + x^r U_r, \tag{4}$$

where

$$U_r = \frac{x}{r+1} + \frac{x^2}{(r+1)(r+2)} + \frac{x^3}{(r+1)(r+2)(r+3)} + \cdots, \tag{5}$$

and this formula also holds for $r = 0$, if one assumes $\Pi(0) = 1$.

If we denote the absolute value of x by ξ, then, by the theorem proved in the introduction that the absolute value of a sum is never greater than the sum of the absolute values of its parts, the absolute value of U_r is certainly not greater than

$$\frac{\xi}{r+1} + \frac{\xi^2}{(r+1)(r+2)} + \frac{\xi^3}{(r+1)(r+2)(r+3)} + \cdots,$$

hence smaller than

$$1 + \frac{\xi}{1} + \frac{\xi^2}{1.2} + \frac{\xi^3}{1.2.3} + \cdots = e^\xi.$$

If therefore we put

$$U_r = q_r e^\xi,$$

then q_r is a function of x about which we establish only so much, that its absolute value is smaller than 1.

Accordingly we write formula (4) for $r = 0, 1, \ldots, n$, and write the terms, ordered in reverse order, in the following survey:

$$\begin{aligned} \Pi(n)e^x &= q_n x^n e^\xi + x^n + nx^{n-1} + n(n-1)x^{n-2} + \cdots + \Pi(n), \\ \Pi(n-1)e^x &= q_{n-1}x^{n-1}e^\xi + x^{n-1} + (n-1)x^{n-2} + (n-1)(n-2)x^{n-3} + \cdots, \\ &\vdots \\ e^x &= q_0 e^\xi + 1. \end{aligned} \tag{6}$$

Now take an arbitrary integral function of degree n in x:

$$f(x) = c_n x^n + c_{n-1}x^{n-1} + c_{n-2}x^{n-2} + \cdots + c_0, \tag{7}$$

and its derivatives

$$\begin{aligned} f'(x) &= nc_n x^{n-1} + (n-1)c_{n-1}x^{n-2} + \cdots, && (1) \\ f''(x) &= n(n-1)c_n x^{n-2} + (n-1)(n-2)c_{n-1}x^{n-3} + \cdots, && (8) \\ &\vdots \\ f^{(n)}(x) &= \Pi(n)c_n, \end{aligned}$$

and put, for abbreviation,

$$F(x) = f(x) + f'(x) + f''(x) + \cdots + f^{(n)}(x). \tag{9}$$

Then $F(x)$ is an integral function of x of degree n. Finally set

$$Q(x) = c_n q_n x^n + c_{n-1}q_{n-1}x^{n-1} + \cdots + c_0 q_0, \tag{10}$$

$$P = c_n\Pi(n) + c_{n-1}\Pi(n-1) + \cdots + c_0, \tag{11}$$

so that $Q(x)$ depends on x, even though it is not expressible rationally by x; but P is independent of x.

If we now multiply the equations (6) in order by $c_n, c_{n-1}, \ldots, c_0$ and add, it follows that

$$e^x P = F(x) + e^\xi Q(x). \tag{12}$$

This formula is now the starting point of the further conclusions.

Suppose that e were an algebraic number. Then there must exist an equation, of degree m:

$$C_0 + C_1 e + C_2 e^2 + \cdots + C_m e^m = 0, \tag{13}$$

whose coefficients $C_0, C_1, \ldots, C_m$ are rational integers, of which C_0 and C_m are different from zero. The point is to show the impossibility of this assumption.

For this purpose we put in equation (12), for x successively, the integers $0, 1, 2, \ldots, m$, so that ξ becomes identical with x, multiply by $C_0, C_1, \ldots, C_m$, and add. Then, by (13), since P is independent of x, there results

$$\begin{aligned} 0 = {} & C_0 F(0) + C_1 F(1) + \cdots + C_m F(m) \\ & + C_0 Q(0) + C_1 e Q(1) + \cdots + C_m e^m Q(m), \end{aligned} \tag{14}$$

and now it is to be proved, from a suitable assumption about the still arbitrary function $f(x)$, that equation (14) is impossible.

We choose a prime number p greater than m[5], and put

$$f(x) = \frac{x^{p-1}(1-x)^p(2-x)^p\cdots(m-x)^p}{\Pi(p-1)}, \tag{15}$$

so that the degree n of $f(x)$ is equal to $(m+1)p-1$, and we now prove two things:

1) $C_0F(0) + C_1F(1) + \cdots + C_mF(m)$ is an integer different from zero, hence, apart from sign, at least equal to 1;

2) $C_0Q(0) + C_1eQ(1) + \cdots + C_me^mQ(m)$ is smaller than 1;

both under the presupposition that p is suitably disposed of. If both of these are proved, then one recognizes the impossibility of equation (14), and hence also that of equation (13), from which (14) had been inferred.

If we order the numerator of $f(x)$ by powers of x, an expression of the form

$$f(x) = \frac{A_{p-1}x^{p-1} + A_px^p + A_{p+1}x^{p+1} + \cdots}{\Pi(p-1)} \tag{16}$$

is obtained, where $A_{p-1}, A_p, A_{p+1}, \ldots$ are integers, and $A_{p-1} = \Pi(m)^p$, and hence is certainly not divisible by p. Therefore, if one compares (16) with the Taylor expansion

$$f(x) = f(0) + xf'(0) + \frac{x^2}{\Pi(2)}f''(0) + \cdots,$$

one has

$$f(0) = f'(0) = f''(0) = \cdots = f^{(p-2)}(0) = 0,$$
$$f^{(p-1)}(0) = A_{p-1}, \quad f^{(p)}(0) = pA_p, \quad f^{(p+1)}(0) = p(p+1)A_{p+1}, \ldots,$$

therefore

$$F(0) = A_{p-1} + pA_p + p(p+1)A_{p+1} + \cdots,$$

an integer not divisible by p. But if we order $f(x)$ by powers of $x-1$, it follows that

$$f(x) = \frac{B_p(x-1)^p + B_{p+1}(x-1)^{p+1} + \cdots}{\Pi(p-1)},$$

where $B_p, B_{p+1}, \ldots$ are again integers. It follows from this, as above, by comparison with

$$f(x) = f(1) + (x-1)f'(1) + \frac{(x-1)^2}{\Pi(2)}f''(1) + \cdots,$$

that

$$F(1) = pB_p + p(p+1)B_{p+1} + \cdots,$$

and hence $F(1)$ is an integer divisible by p, and in exactly the same way one obtains that $F(2), F(3), \ldots, F(m)$ are integers divisible by p. Since one can now also choose p so large that C_0 is not divisible by p, it follows

that $C_0F(0) + C_1F(1) + \cdots + C_mF(m)$ is an integer not divisible by p, hence also not vanishing, and with this 1) is proved.

[5]That there is always a prime number p greater than an arbitrary number μ was already proved by Euclid (Elements, Book IX, no. XX, vol. 2 of Heiberg's edition). The proof is simply that the integer $\Pi(\mu)+1$, which is plainly greater than μ, is divisible by no prime number not greater than μ, since on division by each of the numbers $2, 3, \ldots, \mu$ this number leaves the remainder 1. It is therefore impossible that no prime number lies above μ.

If we can now still show that $Q(x)$, for every positive x, can be made arbitrarily small by increasing p, it follows that, for sufficiently large p, the expression 2) is certainly smaller than 1, and our proof is complete.

But if we go back to the expression (10) for $Q(x)$ and denote by $\gamma_n, \gamma_{n-1}, \ldots, \gamma_0$ the absolute values of the coefficients $c_n, c_{n-1}, \ldots, c_0$ of $f(x)$, then we see, since the $q_n, q_{n-1}, \ldots$ are in absolute value smaller than 1, that for every positive x the absolute value of $Q(x)$ is smaller than

$$\psi(x) = \gamma_n x^n + \gamma_{n-1} x^{n-1} + \cdots + \gamma_0. \tag{17}$$

The coefficients $c_n, c_{n-1}, \ldots, c_0$ of the function $f(x)$ in (15), however, differ from the $\gamma_n, \gamma_{n-1}, \ldots, \gamma_0$ only by sign. Therefore, if we replace x by $-x$, thus form the function

$$f(-x) = \frac{x^{p-1}(1+x)^p(2+x)^p \cdots (m+x)^p}{\Pi(p-1)},$$

this function has the same coefficients as $f(x)$, but all with positive signs. Thus it is no other function than $\psi(x)$.

If we therefore also put

$$X = x(1+x)(2+x)\cdots(m+x),$$

then

$$\psi(x) = \frac{X}{x} \cdot \frac{X^{p-1}}{\Pi(p-1)}, \tag{18}$$

and this approaches, as was proved at the beginning of this paragraph, the limit zero as p grows without bound.

Thus e is a transcendent number.

§. 206.
Transcendence of the Number π.

With the same aids one can now also prove the transcendence of the number π. As definition of this number we use the fact that it is the smallest positive number satisfying the equation

$$e^{i\pi} = -1, \tag{1}$$

when e^x is defined by the formula §. 205, (2).

Suppose then that π, and consequently also $i\pi$, is an algebraic number. Then $i\pi$ is one of the roots of an irreducible rational equation $\chi(x) = 0$ whose coefficients are rational integers.

Let all the roots of this algebraic equation be denoted by $\beta_1, \beta_2, \ldots, \beta_\nu$, and let the coefficient of x^ν in χ be denoted by a; then

$$\chi(x) = a(x-\beta_1)(x-\beta_2)\cdots(x-\beta_\nu), \tag{2}$$

and the products $a\beta_1, a\beta_2, \ldots, a\beta_\nu$ are algebraic integers, hence their integral symmetric functions are rational integers (§. 133).

Since now the number $i\pi$ is to occur among the β, it follows from (1) that

$$(1+e^{\beta_1})(1+e^{\beta_2})\cdots(1+e^{\beta_\nu}) = 0,$$

and if we carry out the multiplication there results an equation

$$1 + \sum e^{\beta_i} + \sum e^{\beta_i+\beta_h} + \sum e^{\beta_i+\beta_h+\beta_k} + \cdots = 0.$$

Among the exponents in this sum the number zero can occur several times; we shall assume $(C-1)$ times, so that C is a positive integer, at least $= 1$. The remaining exponents $\beta_i, \beta_i + \beta_h, \beta_i + \beta_h + \beta_k, \ldots$, which in part may also be equal among themselves, we shall denote by $\alpha_1, \alpha_2, \ldots, \alpha_\mu$, so that the equation

$$C + e^{\alpha_1} + e^{\alpha_2} + \cdots + e^{\alpha_\mu} = 0 \tag{3}$$

holds. Here the $\alpha_1, \alpha_2, \ldots, \alpha_\mu$ are algebraic numbers which, when multiplied by the rational integer a, pass into algebraic integers.

The symmetric functions of all the sums $\beta_i, \beta_i + \beta_h, \beta_i + \beta_h + \beta_k, \ldots$ are at the same time symmetric functions of $\beta_1, \ldots, \beta_\nu$, and hence rational numbers. These sums are consequently also the roots of a rational equation; and since one can separate off the root 0, as often as it is present, the $\alpha_1, \alpha_2, \ldots, \alpha_\mu$ also are the roots of a rational equation; the elementary symmetric functions of $a\alpha_1, a\alpha_2, \ldots, a\alpha_\mu$ are rational integers.

The absolute values of the numbers

$$\alpha_1, \alpha_2, \ldots, \alpha_\mu$$

shall now be denoted by

$$a_1, a_2, \ldots, a_\mu.$$

If we now put in equation (12) of the preceding paragraph $x = 0, \alpha_1, \alpha_2, \ldots, \alpha_\mu$ and apply equation (3), we get

$$\begin{aligned} 0 = {} & CF(0) + F(\alpha_1) + F(\alpha_2) + \cdots + F(\alpha_\mu) \\ & + CQ(0) + e^{\alpha_1}Q(\alpha_1) + e^{\alpha_2}Q(\alpha_2) + \cdots + e^{\alpha_\mu}Q(\alpha_\mu), \end{aligned} \tag{4}$$

and we have to show the impossibility of such an equation by a suitable choice of $f(x)$.

Let p again be a prime number destined to increase without bound, and put

$$f(x) = \frac{a^{\mu p + p - 1} x^{p-1} (x - \alpha_1)^p (x - \alpha_2)^p \cdots (x - \alpha_\mu)^p}{\Pi(p-1)}, \tag{5}$$

so that $f(x)$ is a function with rational coefficients.

We now form, as in the preceding paragraph, by ordering according to powers of x,

$$f(x) = \frac{A_{p-1}x^{p-1} + A_p x^p + A_{p+1}x^{p+1} \cdots}{\Pi(p-1)},$$

where the $A_{p-1}, A_p, \ldots$ are rational integers, and

$$A_{p-1} = (-1)^\mu a^{\mu p + p - 1} \alpha_1^p \alpha_2^p \cdots \alpha_\mu^p.$$

If then we take p greater than each of the rational integers

$$a, \quad a^\mu \alpha_1 \alpha_2 \cdots \alpha_\mu,$$

then A_{p-1} is not divisible by p, and

$$F(0) = A_{p-1} + pA_p + p(p+1)A_{p+1} + \cdots$$

is obtained, that is, an integer not divisible by p. Likewise, if p is taken large enough, C is not divisible by p.

On the other hand, we order the numerator of $f(x)$ according to powers of $a(x - \alpha_1)$ and obtain

$$f(x) = \frac{B_p a^p (x - \alpha_1)^p + B_{p+1} a^{p+1} (x - \alpha_1)^{p+1} + \cdots}{\Pi(p-1)},$$

where $B_p, B_{p+1}, \dots$ are no longer rational, but certainly algebraic integers, since the numerator of $f(x)$ is an integral function of $ax, a\alpha_1, \dots, a\alpha_\mu$.

From this it follows, as in the preceding paragraph, that

$$F(\alpha_1) = pB_p a^p + p(p+1)B_{p+1}a^{p+1} + \cdots .$$

If one forms in the same way $F(\alpha_2), \dots, F(\alpha_\mu)$, and observes that the sum $B_p(\alpha_1) + B_p(\alpha_2) + \cdots + B_p(\alpha_\mu)$ is a rational integer, it follows that

$$F(\alpha_1) + F(\alpha_2) + \cdots + F(\alpha_\mu)$$

is a rational integer divisible by p. From this it finally follows that

$$CF(0) + F(\alpha_1) + F(\alpha_2) + \cdots + F(\alpha_\mu)$$

is a rational integer not divisible by p, and hence also not vanishing, which therefore, apart from sign, is at least $= 1$.

If we can now finally still prove that, under the present assumption about f, the function $Q(x)$ can be made arbitrarily small for every finite x by sufficiently increasing p, then, exactly as above, the impossibility of equation (4) results.

This, however, can be seen as follows: instead of the function

$$f(x) = c_n x^n + c_{n-1}x^{n-1} + \cdots + c_0 \tag{6}$$

we consider the function

$$\psi(x) = \frac{a^{\mu p+p-1}x^{p-1}(x+a_1)^p(x+a_2)^p \cdots (x+a_\mu)^p}{\Pi(p-1)} = \gamma_n x^n + \gamma_{n-1}x^{n-1} + \cdots + \gamma_0,$$

which now has only positive (but not necessarily rational) coefficients. The coefficients $c_n, c_{n-1} \dots$ are formed by multiplication and addition from the numbers $a, -\alpha_1, -\alpha_2, \dots, -\alpha_\mu$, and the corresponding coefficients $\gamma_n, \gamma_{n-1} \dots$ are obtained from them by replacing $-\alpha_1, -\alpha_2, \dots, -\alpha_\mu$ by their absolute values $a_1, a_2, \dots, a_\mu$. From this it follows, by the theorem of the introduction already mentioned above, that the coefficients $\gamma_n, \gamma_{n-1}, \dots$ are certainly not smaller than the absolute values of the corresponding $c_n, c_{n-1} \dots$.

Now for every finite x, whose absolute value is ξ, the absolute value of $Q(x)$ is, by §. 205, (10), smaller, or at least not greater than

$$\gamma_n \xi^n + \gamma_{n-1}\xi^{n-1} + \cdots + \gamma_0 = \psi(\xi),$$

and that $\psi(\xi)$ sinks below every bound if p is large enough follows from the representation

$$\psi(x) = \frac{X}{ax}\frac{X^{p-1}}{\Pi(p-1)},$$

when

$$X = a^{\mu+1}x(x+a_1)(x+a_2)\cdots(x+a_\mu)$$

is put.

Thus it is proved:

The number π is a transcendent number. The "quadrature of the circle" cannot be solved by a geometric construction in which only algebraic curves and surfaces are used.

§. 207.

The General Theorem of Lindemann on the Exponential Function.

The transcendence of the numbers e and π, which has hereby been proved, is contained as a special case in a very general theorem on the exponential function, which Lindemann announced in the paper mentioned above, and a fully carried out proof of which is contained in the cited paper of Weierstrass. We shall prove this theorem here at the close, using the same aids that we have used for the two special cases. The theorem reads as follows:

I. There exists no equation of the form

$$C_1 e^{z_1} + C_2 e^{z_2} + \cdots + C_m e^{z_m} = 0, \tag{1}$$

in which the coefficients $C_1, C_2, \ldots, C_m$ are algebraic numbers and the exponents $z_1, z_2, \ldots, z_m$ are mutually different algebraic numbers, unless all coefficients $C_1, C_2, \ldots, C_m$ are equal to zero.

In order to prove it, we first derive an auxiliary theorem.

Let

$$\alpha_1, \alpha_2, \ldots, \alpha_r$$

be arbitrary real or imaginary, but mutually different, quantities, and let

$$A_1, A_2, \ldots, A_r$$

likewise be arbitrary quantities, not all vanishing. The same is to hold for the two sequences of quantities

$$\beta_1, \beta_2, \ldots, \beta_s, \qquad B_1, B_2, \ldots, B_s.$$

We denote by

$$\gamma_1, \gamma_2, \ldots, \gamma_t$$

the mutually different ones among the rs sums $\alpha_i + \beta_k$, and set

$$A = A_1 e^{\alpha_1} + A_2 e^{\alpha_2} + \cdots + A_r e^{\alpha_r}, \tag{2}$$

$$B = B_1 e^{\beta_1} + B_2 e^{\beta_2} + \cdots + B_s e^{\beta_s},$$

$$AB = C_1 e^{\gamma_1} + C_2 e^{\gamma_2} + \cdots + C_t e^{\gamma_t}. \tag{3}$$

The auxiliary theorem to be proved is then:

1. The coefficients $C_1, C_2, \ldots, C_t$ cannot all vanish.

In the proof of this theorem we can plainly assume that none of the coefficients $A_1, A_2, \ldots, A_r, B_1, B_2, \ldots, B_s$ vanishes (since any vanishing ones can simply be left out).

For the sake of shorter expression, we shall for the moment say, if a, b are two different complex numbers, that a is smaller than b, $(a < b)$, if the real part of a is smaller than the real part of b, or if the real parts are equal and the imaginary part of a is smaller than the imaginary part of b.

If, then, in this sense $a < b$, $b < c$, then also $a < c$, and if $a < b$, $c < d$, then $a + c < b + d$.

Among every finite sequence of mutually different complex numbers there is then a definite smallest one; and if therefore α_1 is the smallest among the numbers α, and β_1 the smallest among the numbers β, then $\alpha_1 + \beta_1$ is the smallest among the numbers γ, and this sum is equal to none of the other sums $\alpha_i + \beta_k$. Thus $C_1 = A_1 B_1$, and C_1 is different from zero, as was to be proved.

This theorem can now be generalized immediately by complete induction.

2. If

$$\begin{aligned} A' &= A'_1 e^{\alpha'_1} + A'_2 e^{\alpha'_2} + \cdots \\ A'' &= A''_1 e^{\alpha''_1} + A''_2 e^{\alpha''_2} + \cdots \\ A''' &= A'''_1 e^{\alpha'''_1} + A'''_2 e^{\alpha'''_2} + \cdots \\ &\cdots \end{aligned}$$

are sums of the form (2) in arbitrary number, and $\gamma_1, \gamma_2, \ldots$ are the mutually different ones among the sums $\alpha'_h + \alpha''_i + \alpha'''_k + \cdots$, then in the product

$$A'A''A''' \cdots = C_1 e^{\gamma_1} + C_2 e^{\gamma_2} + \cdots \tag{4}$$

not all coefficients $C_1, C_2, \ldots$ are equal to zero.

This auxiliary theorem first permits us, in the proof of theorem (1), to make the simplifying assumption that the coefficients $C_1, C_2, \ldots, C_m$ are rational integers.

Namely, if an equation

$$X_1 e^{x_1} + X_2 e^{x_2} + \cdots = 0 \tag{5}$$

with algebraic coefficients $X_1, X_2, \ldots$ and mutually different exponents $x_1, x_2, \ldots$ exists, then we can always assume that the $X_1, X_2, \ldots$ belong to one and the same algebraic field Ω.

We denote by $u_1, u_2, \ldots$ variables and form the norm of the linear form $X_1 u_1 + X_2 u_2 + \cdots$:

$$N(X_1 u_1 + X_2 u_2 + \cdots) = \Phi(u_1, u_2, \ldots). \tag{6}$$

This is an integral homogeneous function of the variables u with rational coefficients, whose degree is equal to the degree of the field Ω. If in (6) we put

$$u_1 = e^{x_1}, \quad u_2 = e^{x_2}, \ldots,$$

then every product of variables u passes into a quantity of the form e^z, where z is a sum of numbers x, and $\Phi(u_1, u_2, \ldots)$ becomes an expression of the form $C_1 e^{z_1} + C_2 e^{z_2} + \cdots$ whose coefficients are rational numbers; if the $X_1, X_2, \ldots$ do not all vanish, then by 2. these coefficients also cannot all be zero after the terms which are equal among the terms of the function $\Phi(e^{x_1}, e^{x_2}, \ldots)$ have been collected into a single term Ce^z. If, however, equation (5) holds, then also $\Phi(e^{x_1}, e^{x_2}, \ldots) = 0$, and consequently

$$C_1 e^{z_1} + C_2 e^{z_2} + \cdots = 0.$$

If among the $C_1, C_2, \ldots$ there are fractional numbers, then by multiplying the whole equation by the principal denominator we can transform them into integers.

3. Thus it remains only to prove that equation (1) cannot exist for any system of rational integers $C_1, C_2, \ldots, C_m$, not all of which vanish.

Accordingly we now assume that there exists an equation

$$A_1 e^{x_1} + A_2 e^{x_2} + \cdots = 0, \tag{7}$$

whose coefficients $A_1, A_2, \ldots$ are rational integers, not all $= 0$, and in which the exponents $x_1, x_2, \ldots$ are mutually different algebraic numbers. We denote by Ω a normal field to which all these numbers $x_1, x_2, \ldots$ belong, and by Θ a primitive number of this field, further by

$$\sigma' = (\Theta, \Theta'), \quad \sigma'' = (\Theta, \Theta''), \ldots$$

the substitutions of the field Ω. If by one of these substitutions, σ', the numbers $x_1, x_2, \ldots$ pass into $x'_1, x'_2, \ldots$, then these also must be mutually different.

Let u now be a variable and

$$U = A_1 e^{ux_1} + A_2 e^{ux_2} + \cdots . \tag{8}$$

Here we carry out all substitutions $\sigma', \sigma'', \ldots$ and denote the functions thereby arising by $U', U'', \ldots$. The product of all these functions, although it is not merely a question of algebraic numbers, we can call the norm of U. This product has the form

$$N(U) = C_1 e^{uz_1} + C_2 e^{uz_2} + \cdots , \tag{9}$$

where the $C_1, C_2, \ldots$ are likewise rational integers. The $z_1, z_2, \ldots$ are numbers of the field Ω, which, if we collect terms with equal exponents into a single term, we may assume to be mutually different. At the same time, because of (7),

$$C_1 e^{z_1} + C_2 e^{z_2} + \cdots = 0. \tag{10}$$

But the expression on the right-hand side of (9) still has, as its origin as a norm shows, the property of remaining unchanged when any one of the substitutions $\sigma', \sigma'', \ldots$ is made. If, however, one expands (9) according to powers of u, this invariance must hold for each term of this series expansion; hence, if h is any positive integer exponent, for

$$C_1 z_1^h + C_2 z_2^h + \cdots .$$

But this sum is a number of the field Ω, and therefore a rational number. We can collect this as follows:

If $g(z)$ denotes any integral function of z with rational coefficients, then

$$C_1 g(z_1) + C_2 g(z_2) + \cdots$$

is a rational number.

4. The proof of theorem I. is hereby reduced to the proof of the following special case: if there exists an equation

$$C_1 e^{z_1} + C_2 e^{z_2} + \cdots + C_m e^{z_m} = 0, \tag{11}$$

in which the $z_1, z_2, \ldots, z_m$ are mutually different algebraic numbers, in which $C_1, C_2, \ldots, C_m$ and the combinations

$$C_1 g(z_1) + C_2 g(z_2) + \cdots + C_m g(z_m) \tag{12}$$

are rational numbers whenever $g(z)$ is any integral function with rational coefficients, then the $C_1, C_2, \ldots, C_m$ must all vanish.

To prove this theorem, first determine a positive rational integer c so that

$$x_1 = cz_1, \quad x_2 = cz_2, \ldots, \quad x_m = cz_m \tag{13}$$

become algebraic integers (§. 133, 5.). We assume the coefficients $C_1, C_2, \ldots, C_m$ to be rational integers, and form an integral function

$$\varphi(x) = ax^r + a_1 x^{r-1} + \cdots + a_r \tag{14}$$

with rational integer coefficients, which has the following properties:

1) The numbers $x_1, x_2, \ldots, x_m$ occur among the roots of $\varphi(x) = 0$.

2) The sum

$$C_1\varphi'(x_1) + C_2\varphi'(x_2) + \cdots + C_m\varphi'(x_m) = k,$$

which by the assumptions (12), (13) is a rational integer, is different from zero.

To see that such a function $\varphi(x)$ always exists, one first takes a function $\chi(x)$ which satisfies only condition 1), and the second condition that none of the numbers $x_1, x_2, \ldots, x_m$ is a double root of $\chi(x)$. Such a function plainly exists [for example, in order to form a function χ, one can free the norm of the product $(x - x_1)(x - x_2)\cdots(x - x_m)$ from common divisors with its derivatives]. If one then sets

$$\varphi(x) = x^h\chi(x), \quad \varphi'(x_1) = x_1^h\chi'(x_1), \quad \varphi'(x_2) = x_2^h\chi'(x_2), \ldots,$$

then the sum

$$C_1\varphi'(x_1) + \cdots + C_m\varphi'(x_m) = C_1\chi'(x_1)x_1^h + \cdots + C_m\chi'(x_m)x_m^h$$

certainly cannot vanish for every exponent h, since otherwise, contrary to the assumption, $C_1\chi'(x_1)\ldots C_m\chi'(x_m)$ would all have to be equal to zero.

After the existence of a function $\varphi(x)$ as required has thus been established, we apply formula §. 205, (12):

$$e^x P(x) = F(x) + e^\xi Q(x).$$

In it we put $x = z_1, z_2, \ldots, z_m$, and denote the absolute values of $z_1, z_2, \ldots, z_m$ by $\xi_1, \xi_2, \ldots, \xi_m$.

Then, by (11), there results

$$\begin{aligned} 0 &= C_1F(z_1) + C_2F(z_2) + \cdots + C_mF(z_m) \\ &\quad + C_1e^{\xi_1}Q(z_1) + C_2e^{\xi_2}Q(z_2) + \cdots + C_me^{\xi_m}Q(z_m), \end{aligned} \tag{15}$$

and, if $f(x)$ is an arbitrary integral function of degree n,

$$F(z) = f(z) + f'(z) + \cdots + f^{(n)}(z). \tag{16}$$

Putting, with p denoting a natural prime number which can be taken arbitrarily large, $x = cz$ [by (13)] and

$$f(z) = \frac{\varphi(x)^{p-1}\varphi'(x)}{\Pi(p-1)}, \tag{17}$$

where $\varphi(x)$ is the function satisfying conditions 1), 2), we get the required choice of f.

By ordering according to powers of $(x - x_1)$ let there result

$$\varphi(x)^{p-1}\varphi'(x) = \varphi'(x_1)^p(x - x_1)^{p-1} + A_p(x_1)(x - x_1)^p + A_{p+1}(x_1)(x - x_1)^{p+1} + \cdots,$$

and in this the $A_p(x_1), A_{p+1}(x_1), \ldots$ are algebraic integers, indeed rational functions of x_1. On the other hand, by Taylor's theorem,

$$f(z) = f(z_1) + (z - z_1)f'(z_1) + \frac{(z - z_1)^2}{1.2}f''(z_1) + \cdots$$

and $c(z - z_1) = x - x_1$. The comparison then gives

$$f(z_1) = 0, \quad f'(z_1) = 0, \ldots, \quad f^{(p-2)}(z_1) = 0,$$

$$f^{(p-1)}(z_1) = c^{p-1}\varphi'(x_1)^p, \quad f^{(p)}(z_1) = pc^pA_p(x_1), \quad f^{(p+1)}(z_1) = p(p+1)c^{p+1}A_{p+1}(x_1), \ldots,$$

and consequently

$$F(z_1) = c^{p-1}\varphi'(x_1)^p + pc^pA_p(x_1) + p(p+1)c^{p+1}A_{p+1}(x_1) + \cdots;$$

here x_1, z_1 can be replaced by x_2, z_2 or by x_3, z_3, and so on.

Accordingly the rational integer

$$C_1F(z_1) + C_2F(z_2) + \cdots + C_mF(z_m)$$

is congruent modulo p to

$$c^{p-1}\left[C_1\varphi'(x_1)^p + C_2\varphi'(x_2)^p + \cdots + C_m\varphi'(x_m)^p\right].$$

Since now $C_1, C_2, \ldots, C_m$ are rational integers, one has $C_1^p \equiv C_1$, $C_2^p \equiv C_2, \ldots \pmod p$, and by application of the polynomial theorem there results

$$\begin{aligned} C_1\varphi'(x_1)^p + C_2\varphi'(x_2)^p + \cdots + C_m\varphi'(x_m)^p &\equiv \left[C_1\varphi'(x_1) + C_2\varphi'(x_2) + \cdots + C_m\varphi'(x_m)\right]^p \\ &\equiv k^p \pmod p. \end{aligned}$$

Thus we obtain the congruence

$$C_1F(z_1) + C_2F(z_2) + \cdots + C_mF(z_m) \equiv c^{p-1}k^p \pmod p. \tag{18}$$

Now the numbers c, k are independent of p, and we can therefore assume p so large that it divides neither c nor k.

5. Then the sum $C_1F(z_1) + C_2F(z_2) + \cdots + C_mF(z_m)$ is a rational integer different from zero, hence in absolute value at least equal to 1.

We now denote by $\varphi_1(x)$ the integral function which is obtained from $\varphi(x)$ by replacing the negative ones among the coefficients $a, a_1, a_2, \ldots$ by their positive values. The function

$$f_1(z) = \frac{\varphi_1(x)^{p-1}\varphi_1'(x)}{\Pi(p-1)},$$

which is obtained according to (17), then has only positive coefficients, and these coefficients are in absolute value certainly not smaller than the corresponding coefficients of $f(z)$, because the coefficients of $f_1(z)$ arise from the same numbers by addition as those which, in forming the coefficients of $f(z)$, are partly added and partly subtracted.

From this, however, by formula §. 205, (10), if we denote by ξ the absolute value of x, for the absolute value of $Q(z)$ we get

$$|Q(z)| \leq \frac{\varphi_1(\xi)^{p-1}\varphi_1'(\xi)}{\Pi(p-1)}.$$

6. and therefore $Q(z)$ can be made arbitrarily small for every finite z by sufficiently increasing p.

By 5. and 6. it is now shown that, if p is large enough, equation (15) cannot hold. Thus 4., and with it the whole theorem I., is proved.

This theorem now allows many applications. It gives us first the transcendence of e, if we take $C_1, C_2, \ldots, z_1, z_2, \ldots$ as rational integers. It further gives the transcendence of π. For from the equation $1 + e^{i\pi} = 0$ it follows by I. that $i\pi$, and consequently also π, cannot be algebraic.

It follows further from it:

For every algebraic number x, with the exception of $x = 0$, $X = e^x$ is a transcendent number.

For every algebraic X, with the exception of $X = 1$, every natural logarithm $x = \log X$ is a transcendent number.

For every arc which stands to the radius in an algebraically expressible ratio x, with the exception of $x = 0$, $X = \sin x$ is a transcendent number.

This follows by I. from $2iX = e^{ix} - e^{-ix}$.

The same holds for the other trigonometric lines, $\cos x$, $\operatorname{tg} x$, and for the chord $\frac{1}{2}\sin\frac{x}{2}$. To mention one more: the transcendent equation $\operatorname{tg} x = ax$ has, for algebraic a different from 0, only transcendent numbers as roots.